

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

```
In [4]: import pandas as pd
df=pd.read_csv("Flight_Booking.csv")
df=df.drop(columns=["Unnamed: 0"])
df.head()
```

```
Out[4]:
```

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai
4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai

```
In [3]: df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300153 entries, 0 to 300152
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   airline                300153 non-null object
1   flight                 300153 non-null object
2   source_city            300153 non-null object
3   departure_time         300153 non-null object
4   stops                  300153 non-null object
5   arrival_time           300153 non-null object
6   destination_city       300153 non-null object
7   class                  300153 non-null object
8   duration                300153 non-null float64
9   days_left              300153 non-null int64
10  price                   300153 non-null int64
dtypes: float64(1), int64(2), object(8)
memory usage: 25.2+ MB
```

Out[3]:

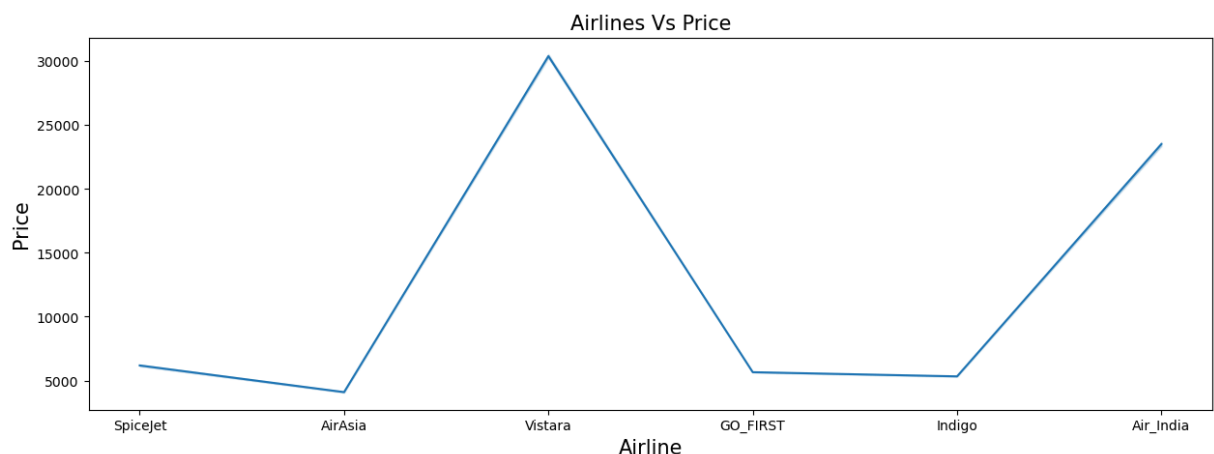
	duration	days_left	price
count	300153.000000	300153.000000	300153.000000
mean	12.221021	26.004751	20889.660523
std	7.191997	13.561004	22697.767366
min	0.830000	1.000000	1105.000000
25%	6.830000	15.000000	4783.000000
50%	11.250000	26.000000	7425.000000
75%	16.170000	38.000000	42521.000000
max	49.830000	49.000000	123071.000000

In [8]: `df.isnull().sum()`

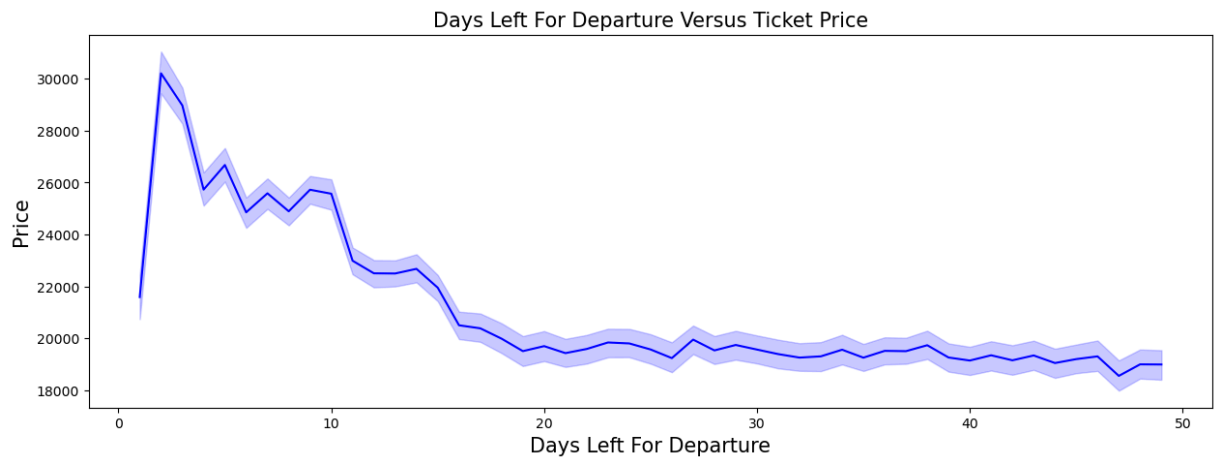
```
Out[8]: airline          0
flight                0
source_city          0
departure_time       0
stops                0
arrival_time         0
destination_city     0
class                0
duration             0
days_left           0
price                0
dtype: int64
```

```
In [10]: plt.figure(figsize=(15,5))
sns.lineplot(x=df['airline'],y=df['price'])
plt.title('Airlines Vs Price',fontsize=15)
plt.xlabel('Airline',fontsize=15)
plt.ylabel('Price',fontsize=15)
plt.show
```

Out[10]: `<function matplotlib.pyplot.show(close=None, block=None)>`

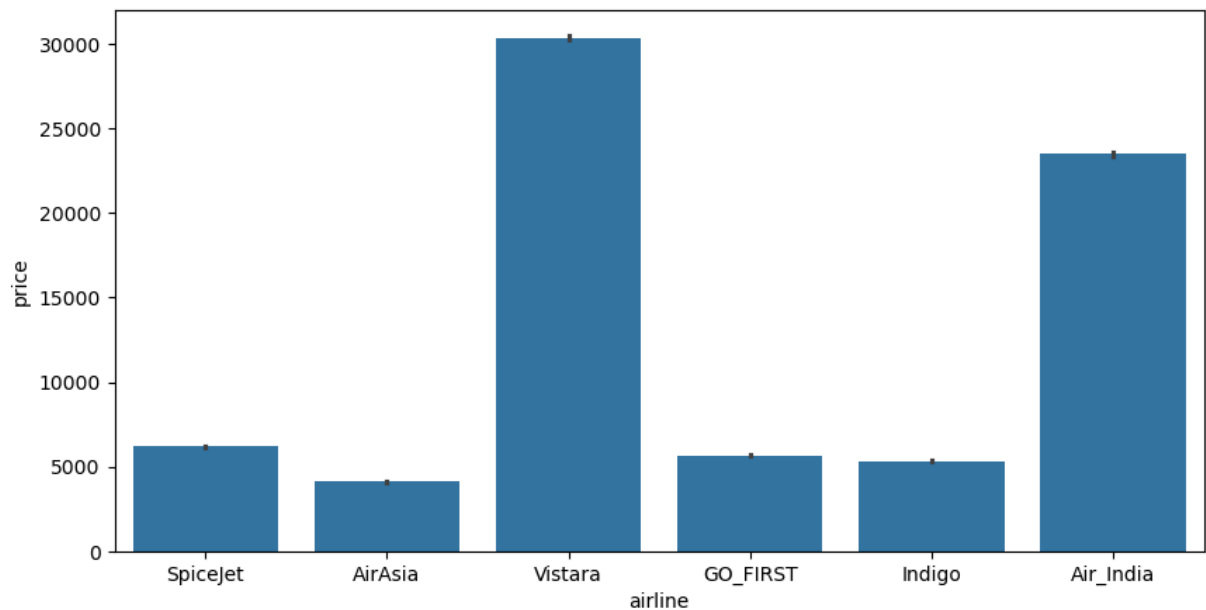


```
In [12]: plt.figure(figsize=(15,5))
sns.lineplot(data=df,x='days_left',y='price',color='blue')
plt.title('Days Left For Departure Versus Ticket Price',fontsize=15)
plt.xlabel('Days Left For Departure',fontsize=15)
plt.ylabel('Price',fontsize=15)
plt.show()
```



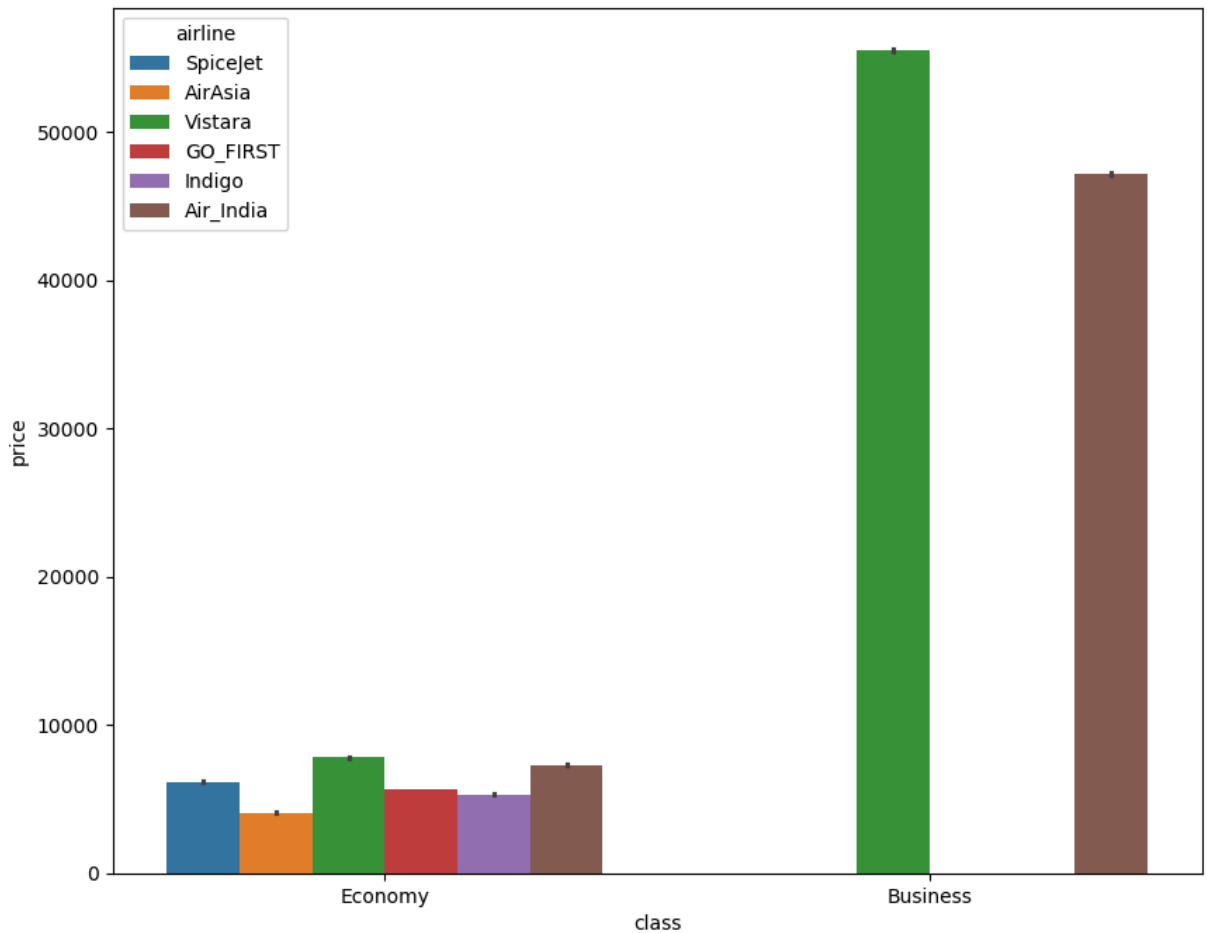
```
In [14]: plt.figure(figsize=(10,5));
sns.barplot(x='airline',y='price',data=df)
```

Out[14]: <Axes: xlabel='airline', ylabel='price'>

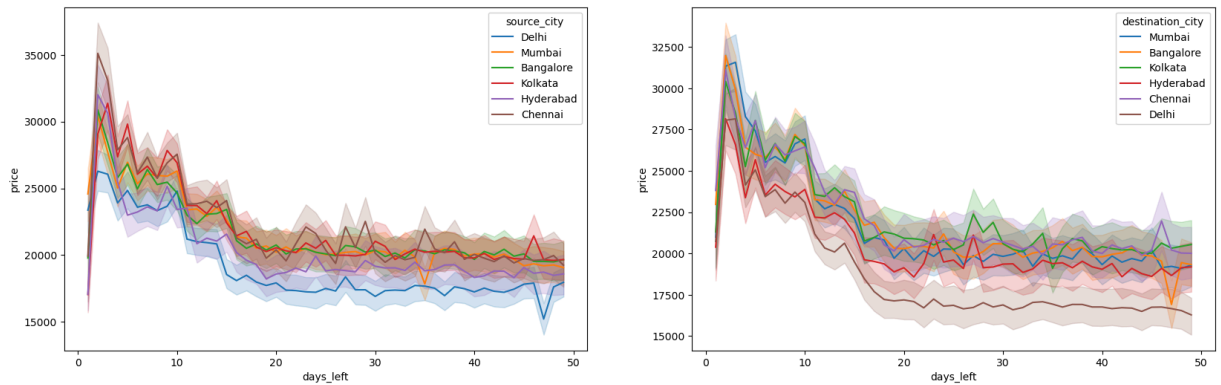


```
In [16]: plt.figure(figsize=(10,8));
sns.barplot(x='class',y='price',data=df,hue='airline')
```

Out[16]: <Axes: xlabel='class', ylabel='price'>



```
In [18]: fig,ax=plt.subplots(1,2,figsize=(20,6))
sns.lineplot(x='days_left',y='price',data=df,hue='source_city',ax=ax[0])
sns.lineplot(x='days_left',y='price',data=df,hue='destination_city',ax=ax[1])
plt.show()
```



```
In [20]: plt.figure(figsize=(15,23))

plt.subplot(4, 2, 1)
sns.countplot(x=df["airline"], data=df)
plt.title("Frequency of Airline")

plt.subplot(4, 2, 2)
sns.countplot(x=df["source_city"], data=df)
plt.title("Frequency of Source City")
```

```
plt.subplot(4, 2, 3)
sns.countplot(x=df["departure_time"], data=df)
plt.title("Frequency of Departure Time")

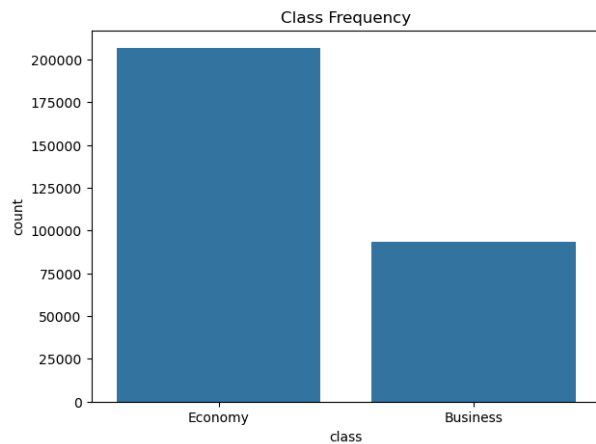
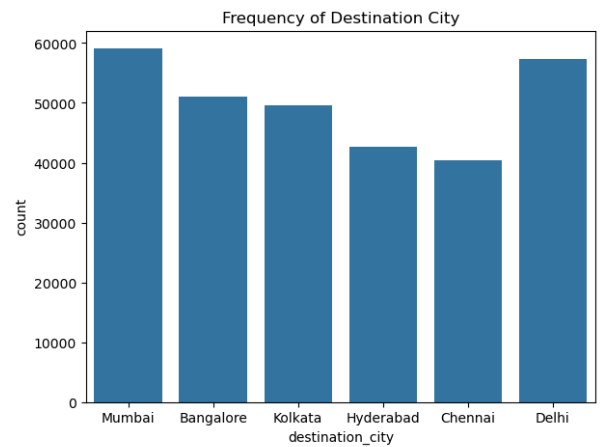
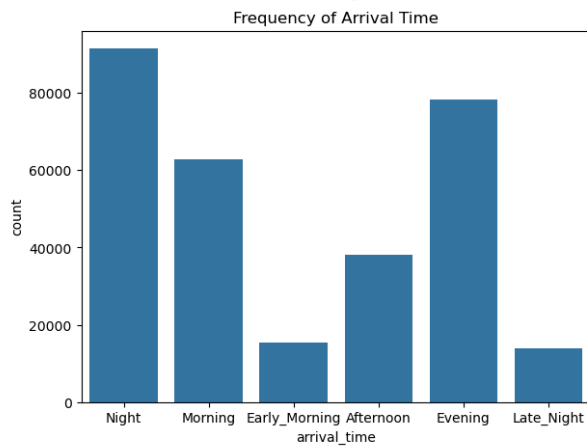
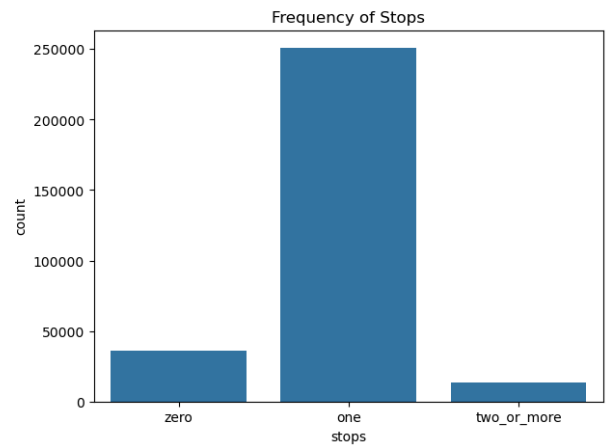
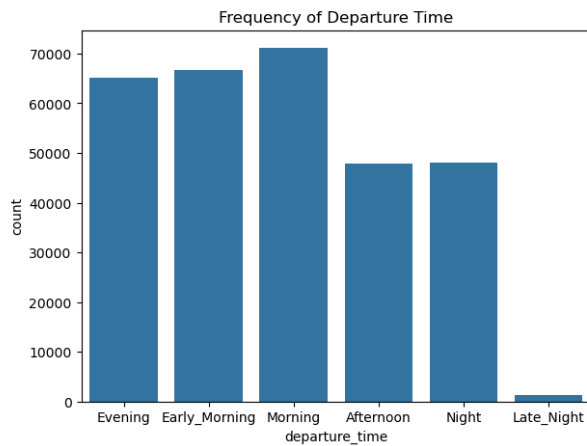
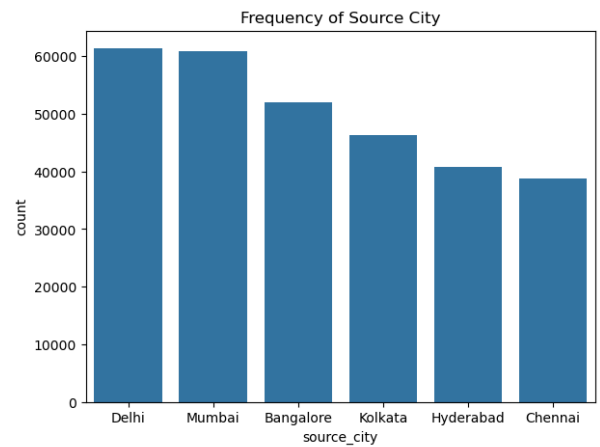
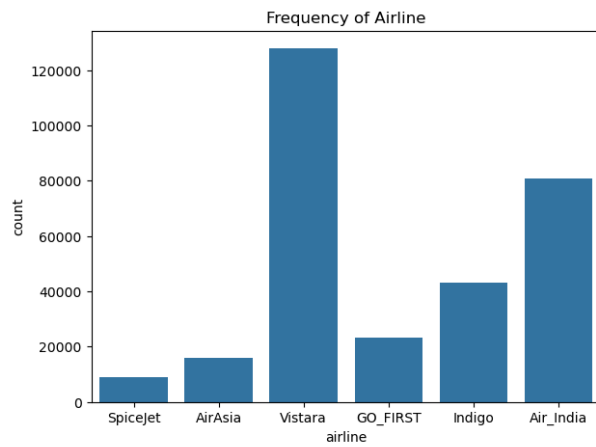
plt.subplot(4, 2, 4)
sns.countplot(x=df["stops"], data=df)
plt.title("Frequency of Stops")

plt.subplot(4, 2, 5)
sns.countplot(x=df["arrival_time"], data=df)
plt.title("Frequency of Arrival Time")

plt.subplot(4, 2, 6)
sns.countplot(x=df["destination_city"], data=df)
plt.title("Frequency of Destination City")

plt.subplot(4, 2, 7)
sns.countplot(x=df["class"], data=df)
plt.title("Class Frequency")

plt.show()
```



```
In [22]: from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
```

```

df["airline"]=le.fit_transform(df["airline"])
df["source_city"]=le.fit_transform(df["source_city"])
df["departure_time"]=le.fit_transform(df["departure_time"])
df["stops"]=le.fit_transform(df["stops"])
df["arrival_time"]=le.fit_transform(df["arrival_time"])
df["destination_city"]=le.fit_transform(df["destination_city"])
df["class"]=le.fit_transform(df["class"])
df.info()

```

```

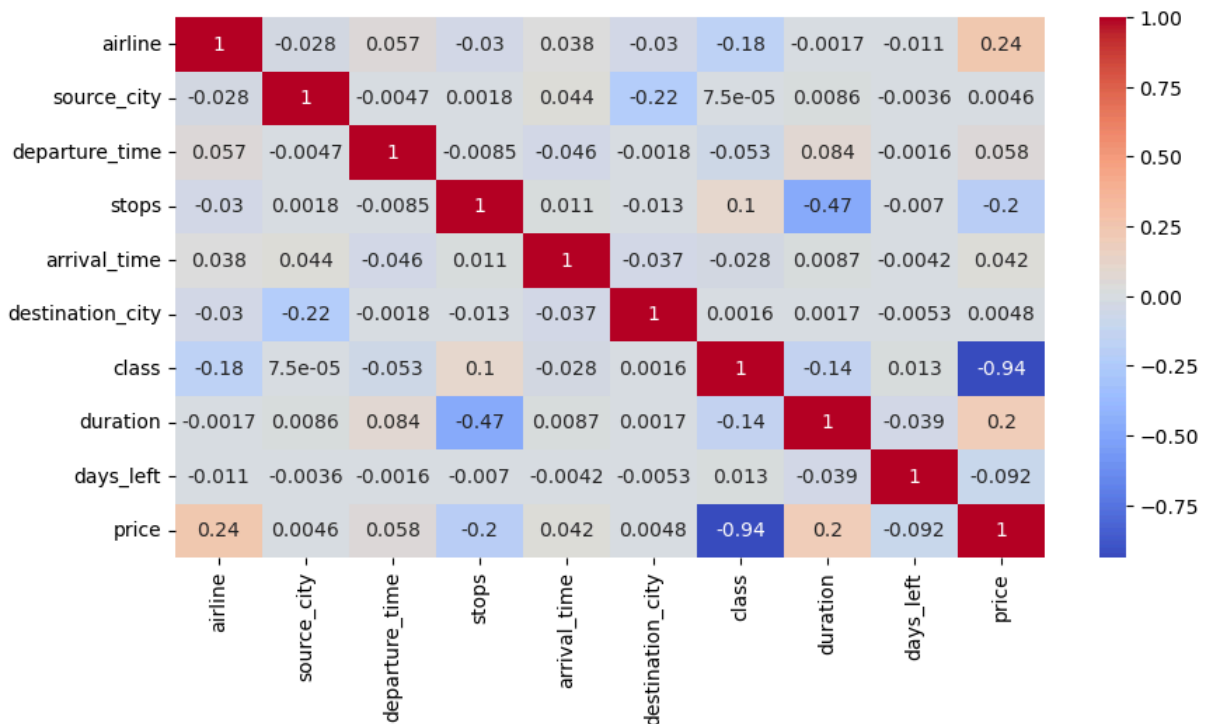
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300153 entries, 0 to 300152
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   airline                300153 non-null  int64
1   flight                 300153 non-null  object
2   source_city            300153 non-null  int64
3   departure_time         300153 non-null  int64
4   stops                  300153 non-null  int64
5   arrival_time           300153 non-null  int64
6   destination_city       300153 non-null  int64
7   class                  300153 non-null  int64
8   duration                300153 non-null  float64
9   days_left              300153 non-null  int64
10  price                  300153 non-null  int64
dtypes: float64(1), int64(9), object(1)
memory usage: 25.2+ MB

```

```

In [26]: plt.figure(figsize=(10, 5))
sns.heatmap(df.select_dtypes(include='number').corr(),
            annot=True,
            cmap='coolwarm')
plt.show()

```



```
In [28]: from statsmodels.stats.outliers_influence import variance_inflation_factor
col_list = []
for col in df.columns:
    if ((df[col].dtype != 'object') & (col != 'price')):
        col_list.append(col)

x = df[col_list]
vif_data = pd.DataFrame()
vif_data["feature"] = x.columns
vif_data["VIF"] = [variance_inflation_factor(x.values, i)
                    for i in range(len(x.columns))]
print(vif_data)
```

	feature	VIF
0	airline	3.393124
1	source_city	2.927766
2	departure_time	2.779427
3	stops	1.426614
4	arrival_time	3.684550
5	destination_city	2.885337
6	class	2.849370
7	duration	4.113876
8	days_left	3.976790

```
In [30]: df=df.drop(columns=["stops"])

from statsmodels.stats.outliers_influence import variance_inflation_factor
col_list = []
for col in df.columns:
    if ((df[col].dtype != 'object') & (col != 'price')):
        col_list.append(col)

X = df[col_list]
vif_data = pd.DataFrame()
vif_data["feature"] = X.columns
vif_data["VIF"] = [variance_inflation_factor(X.values, i)
                    for i in range(len(X.columns))]
print(vif_data)
```

	feature	VIF
0	airline	3.370020
1	source_city	2.895803
2	departure_time	2.746255
3	arrival_time	3.632792
4	destination_city	2.857808
5	class	2.776721
6	duration	3.429344
7	days_left	3.950132